

MINQIN, China

WMO: 526810

Lat: 38.6320N

Lon: 103.0890E

Elev: 4485

StdP: 12.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	3.1	-17.7	2.5	13.4	-14.5	2.9	14.8	20.0	29.3	17.3	28.8	3.2	90	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	93.4	62.9	90.6	61.7	88.0	60.7	67.5	82.9	65.8	80.8	64.4	79.6	7.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	102.5	72.4	61.2	95.5	71.4	59.5	89.8	70.5	34.8	83.6	33.3	80.7	32.1	79.8	77.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.1	16.8	14.1	-6.4	99.7	4.6	3.7	-9.7	102.3	-12.4	104.5	-15.0	106.6	-18.3	109.2	
			DB												
			WB	-7.4	70.0	4.3	2.8	-10.5	72.0	-13.0	73.6	-15.4	75.1	-18.5	77.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	18.7	26.9	39.8	53.7	63.8	72.2	75.7	73.0	63.1	49.9	35.2	22.5
	DBStd	20.95	7.80	8.63	8.66	8.27	6.80	5.54	5.62	5.45	6.21	7.11	7.48	7.28
	HDD50	3391	971	646	329	56	3	0	0	0	1	88	446	851
	HDD65	6535	1436	1066	780	347	103	10	2	6	106	468	895	1316
	CDD50	3266	0	0	14	167	430	665	796	714	393	86	1	0
	CDD65	937	0	0	0	8	66	226	333	255	48	1	0	0
	CDH74	9807	0	0	2	134	821	2387	3531	2425	485	22	0	0
	CDH80	3857	0	0	0	25	216	926	1637	947	105	1	0	0
Wind	WSAvg	6.0	4.9	5.6	6.6	7.4	6.9	6.4	6.2	6.0	5.3	5.3	5.5	5.3
Precipitation	PrecAvg	4.4	0.0	0.0	0.1	0.2	0.4	0.6	0.8	1.2	0.6	0.3	0.1	0.0
	PrecMax	7.3	0.1	0.2	0.9	0.7	1.6	2.4	2.9	3.5	2.5	1.3	0.9	0.1
	PrecMin	1.7	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0
	PrecStd	1.4	0.0	0.0	0.2	0.2	0.4	0.5	0.6	0.9	0.5	0.3	0.2	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	56.2	70.7	83.5	89.0	93.9	98.8	94.9	87.7	76.9	61.9	48.7
		MCWB	32.3	38.2	45.0	51.8	55.3	60.4	65.6	64.5	59.4	51.6	41.8	33.4
	2%	DB	40.3	49.9	64.5	77.8	85.5	90.7	94.8	91.4	82.9	71.9	57.3	42.8
		MCWB	28.6	34.6	42.3	50.0	54.8	59.5	63.9	63.1	57.0	50.1	40.2	30.4
	5%	DB	35.6	45.6	60.1	73.4	82.0	88.3	91.8	88.5	79.6	67.8	53.1	38.6
		MCWB	25.8	32.3	40.5	48.0	53.4	58.8	63.0	62.1	56.2	48.2	37.9	28.2
	10%	DB	31.6	41.6	55.7	69.3	78.7	85.6	88.8	85.5	76.2	63.9	48.8	34.8
		MCWB	23.3	30.2	38.3	46.2	52.0	57.9	62.4	61.2	55.6	46.4	36.0	26.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	38.7	46.2	54.2	58.9	65.2	70.8	69.8	64.5	54.9	43.5	33.9
		MCDB	46.0	53.9	65.9	74.4	79.6	82.3	91.1	80.8	72.2	69.6	57.9	47.2
	2%	WB	28.7	35.3	43.4	51.4	56.7	63.5	68.1	67.5	61.7	52.2	41.1	30.8
		MCDB	39.7	48.6	62.6	73.3	79.0	81.0	85.2	81.7	73.8	65.9	55.3	42.2
	5%	WB	25.9	32.7	41.2	49.4	55.1	61.9	66.3	66.0	59.5	50.1	38.7	28.4
		MCDB	35.1	44.6	58.4	69.7	76.9	78.9	82.0	80.2	71.0	64.7	51.3	38.2
	10%	WB	23.5	30.4	39.0	47.3	53.5	60.4	64.8	64.4	58.0	48.0	36.6	26.2
		MCDB	31.3	41.1	54.8	66.6	74.2	77.8	80.4	79.0	70.2	61.5	48.2	34.3
Mean Daily Temperature Range	5% DB	MDBR	22.1	23.5	23.5	23.2	22.7	21.7	20.8	20.6	20.8	23.2	22.0	21.0
		MCDBR	26.7	28.7	29.6	28.8	27.5	26.0	25.6	25.1	26.7	28.6	27.8	25.0
		MCWBR	17.9	17.4	15.0	12.6	10.3	8.8	7.8	7.8	8.9	12.1	15.2	16.1
	5% WB	MCDBR	26.3	27.3	27.8	25.3	23.6	20.8	20.2	20.5	20.0	25.4	25.8	24.3
		MCWBR	17.8	16.7	14.4	11.4	9.5	7.6	7.3	7.2	8.1	11.7	14.5	15.8
Clear-Sky Solar Irradiance	taub	0.338	0.394	0.458	0.501	0.504	0.497	0.487	0.484	0.466	0.420	0.360	0.330	
	taud	2.066	1.933	1.796	1.736	1.791	1.871	1.970	1.961	1.935	1.966	2.053	2.087	
	Ebn at Noon	255	254	249	248	250	252	253	250	245	242	244	248	
	Edh at Noon	37	49	63	71	69	63	57	56	54	47	38	34	
All-Sky Solar Radiation	RadAvg	881	1196	1594	1927	2147	2160	2021	1846	1511	1309	998	826	
	RadStd	89	62	91	95	89	135	121	115	119	73	60	56	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	N/A	N/A	+1.31	N/A	N/A	N/A	N/A	+223	+117
(47)	Regional (1 neighbor)	+0.65	N/A	N/A	+1.04	N/A	N/A	N/A	N/A	+223	+117

Nomenclature: See separate page